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RESEARCH-ARTICLE

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Engagement Intensity Prediction with Facial Behavior Features

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ABSTRACT

This paper describes an approach for the engagement prediction task, a sub-challenge of the 7th Emotion Recognition in the Wild Challenge (EmotiW 2019). Our method involves three fundamental steps: feature extraction, regression and model ensemble. In the first step, an input video is divided into multiple overlapped segments (instances) and the features extracted for each instance. The combinations of Long short-term memory (LSTM) and Fully connected layers deployed to capture the temporal information and regress the engagement intensity for the features in previous step. In the last step, we performed fusions to achieve better performance. Finally, our approach achieved a mean square error of 0.0597, which is 4.63% lower than the best results last year.

CCS CONCEPTS

• **Computing methodologies → Activity recognition and understanding**.

KEYWORDS

Engagement detection, E-learning Environment, Ensemble model, Facial behavior, Affective computing

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1 INTRODUCTION

Automatic user engagement prediction is an active research area that aims to predict the engagement of the connection between two or more instances. It attracted the attention of many researchers because of its applications in many fields such as education, human-computer interaction, social interaction, as well as health care. In the learning environment, engagement means student engagement which organized into *behavioral* - enthusiasm for participating in the learning process and school-related activities; *emotional* involves the affective states in the learning task or learning context including interest and boredom; *cognitive* refers to the investment in learning [8]. The traditional classroom has fundamentally transformed into the e-learning environment such as Massive Online Open Courses (MOOC), which takes advantage of new technologies to transfer the knowledge efficiently. In this situation, we need to explore new approaches to measure the engagement level during learning time toward improving the course quality.

The effects of feature types (Box Filter, Energy Filters features) and classifier combinations from the automatic facial expression to predict the student engagement studied in [15]. The dataset with four levels of engagement collected while subjects were playing cognitive skills training software on an iPad. The author in [4] analyzed 19 Action Units (AUs), head pose, and position as well as the body movement. Dataset with five affective state levels collected in a school computer lab while students were interacting with educational game.

In [12], the facial features and heart rate employed to detect the engagement while students completed a structured writing activity. They analyzed the Animation Units from (ANUs) Kinect Face tracker, Local Binary Patterns in Three Orthogonal Planes (LBP-TOP), and heart rate features. The overall best results were found using a fusion of the best two channels (LBP-TOP, heart rate).

An "in the wild" dataset with four engagement intensities was collected [11]. They fed the fused features of Head Pose and Eye Gaze to a deep multi-instance network architecture to regress the intensity value. [13] proposed a set of Eye Gaze, AUs, Head Pose features with Gated Recurrent Unit networks, while [14] fed a set of AU intensities features to a Temporal Convolution Networks. In another way, [17] performed a late fusion of Eye Gaze, Head Pose with Body Posture. In [6], authors made the ensembles of the traditional machine learning method (K-mean, AdaBoost), deep learning method (Bidirectional LSTM) and rules-based method.

In this paper, we focus on analyzing facial behavior features for engagement prediction with the experiments on the dataset provided in [11]. Our paper organized as follows: Section 2 describes the feature extraction and the network architecture, then experiments and results show in Section 3, Section 4 summarizes conclusions and the future works.

2 PROPOSED METHOD

In our approach, each video goes through OpenFace [3], a well-known toolkit for facial behavior analysis in the computer vision community, which implemented MTCNN [19] for face detection, [2, 18] for facial landmark detection and tracking, [16] for eye gaze estimation, and [1] for facial action unit detection. Based on the face region extracted by OpenFace, the facial feature obtained by a pre-trained SE-ResNet-50 model [5, 9, 10]. We divided the video sequence v into ℓ segments $s_1, s_2, \ldots, s_\ell$ with $|s_i \cap s_{i+1}| = r$ and $|s_i = s_{i+1}|$, $i = \overline{1, \ell - 1}$. The features for each segment obtained based on the statistical characteristics of OpenFace features and SE-ResNet feature on frames belong to them. 1 shows our overall system pipeline.

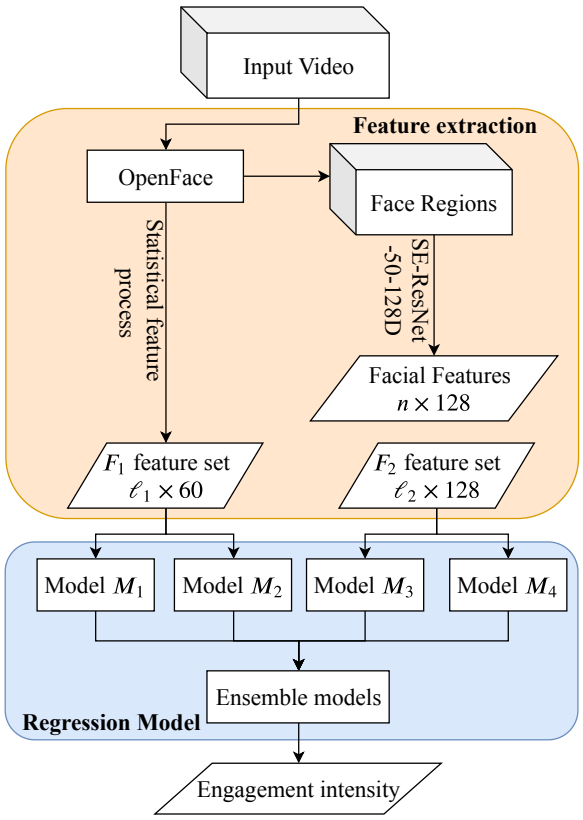

Figure 1: The overall of our system pipeline.

Feature extraction

Previous works in engagement prediction showed that the engagement had a highly correlated with the movement in the face (eye gaze, head pose) [6, 11, 13, 17] as well as the facial expression (LBP-TOP, Gabor features) [11, 12, 15]. In this paper, we consider the combination of eye gaze, head pose features with the automatic feature extracted by convolutional neural networks.

Eye gaze and head pose features. OpenFace provides 56 landmarks to determine eye regions in each frame (2D landmarks in pixels) as well as in the real world (3D landmarks in millimeters). The eye gaze vector for each eye as well as gaze angle in radian for both eyes were estimated. We deployed this information to get the F_1 feature set.

To capture the changing of head pose and gaze direction, in each segment, we calculated the mean and standard deviation of head location, rotation, and gaze vector of both eyes as well as gaze angle. In additional, we observed that the eye shape - the y distance between two eye landmark points whose the x distance is minimum, and eye locations in the frame as well as in the real world, are one of the reasons that makes the change in the gaze direction. These properties were observed based on the mean, standard deviation, and the ratio between the minimum and maximum values in each segment. 1 shows the dimension of each feature type in F_1.

Facial features. The feature set F_2 was obtained by feeding the face regions into a CNN model. These faces were achieved from 68 landmarks, providing by Open Face. Before inputting to the network, the face images were resized so that the shorter size was 256 pixels and then were cropped out the center 224×224 portion of the image.

Table 1: A summary of F_1 feature set extracted based on OpenFace.

Feature type	Feature information	Dim
Gaze direction	mean, standard deviation	16
Eye landmarks 2D and 3D, distance from eye to camera	mean, standard deviation, coefficient of variation, ratio between the min and max values	32
Head pose	mean, standard deviation	12

The facial features were obtained by SE-ResNet-50 [5, 9, 10] which integrated the Squeeze-and-Excitation (SE) blocks, that adaptively recalibrated channel-wise feature responses by explicitly modeling the relationship between the channels of its convolutional feature, into ResNet-50. The efficiency of SE blocks demonstrated for object and scene classification by winning in the ILSVRC 2017 classification competition. To take advantage of SE blocks and existing state-of-the-art deep architectures, we used SE-ResNet-50 which trained on MS-Celeb-1M [9], a large scale real-world face image dataset, and then fine-tune on VGGFace2 dataset, a large-scale face recognition dataset with a large variation in pose, age, illumination, ethnicity, and profession. The authors in [5] performed the training and fine-tune.

Engagement intensity regression models

For the model architecture, we deployed two different networks A_1, A_2 for each feature set. One model is LSTM-FC, learning in the original feature set, while the other is extended by using the fully connected right before LSTM (FC-LSTM-FC), learning in the new representation of the data, as shown in 2. We used ReLU activation in each FC layer and tanh function in LSTM layers. In the video, the engagement varied across time depending on the current learning content as well as the previous. To capture these variations, we used the 2 LSTM layers followed by 2 FC layers to learn, obtain the relationships between consecutive segments, and make the prediction for each segment produced by the last FC layer. These values were passed to a pooling layer which took the average of them and outputted the engagement intensity for the whole video. In this work, we took a multi-level ensemble based on the results from every single model to improve predictions. These improvements were due to the diversity in the feature type as well as feature representation.

3 EXPERIMENT RESULTS

Dataset

In this paper, we experimented with an "Engagement in the wild dataset" [11] which used in Engagement prediction task, a sub-challenge in EmotiW 2019 [7]. The data was recorded with a webcam on a laptop or computer, a mobile phone camera while the student participants were watching five minutes long MOOC video. The environment in which students watched learning material also different from the computer lab, canteen, playground to hostel rooms. The dataset included 91 subjects (27 females and 64 males) with 147, 48 and 67 videos, each approximately 5 minutes long, for training, validation, and testing, respectively. Four levels of engagement $\{0, 1, 2, 3\}$ were labeled by annotators. 3 shows the distribution of the development data. In this task, the problem was formulated as an regression problem with the output in $[0, 1]$ corresponding to 4 engagement levels $(0 : 0, 0.33 : 1, 0.66 : 2, 1 : 3)$. The system performance evaluated with mean square error (MSE) between the ground truth, and the predicted value of the test set.

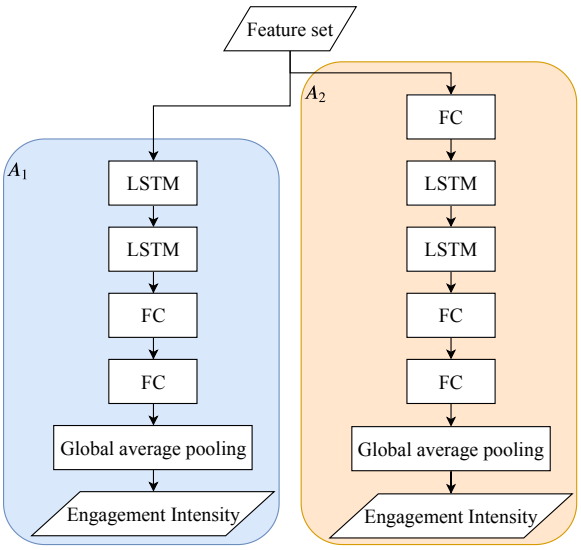

Figure 2: The pipeline of A_1, A_2.

Experiment results

In the experiments, we empirically selected $\ell_1 = 15$ and $\ell_2 = 21$ which are the number of segments for F_1 and F_2, respectively. We have two different models for each feature set, as shown in 1. Our model trained on the training set and adjusted parameters with the validation set without any data augmentation as well as resampling on the training set. 2 shows the detail of each model. The results illustrate that A_1 achieved better MSE than A_2 for both feature set F_1, F_2.

To improve the overall performance, we made the late fusion of models and evaluated the test set through five submissions. The fusion was considered based on the following

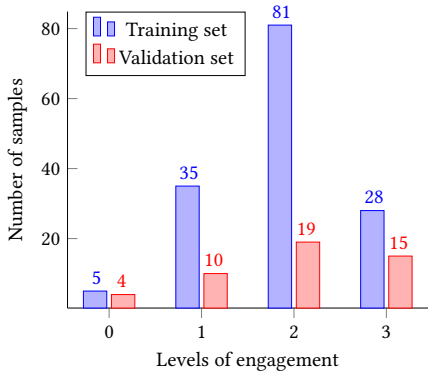

Figure 3: The class wise distribution of training and validation data.

Table 2: The details of regression models. Dimensions are the output shape of FC, LSTM layers in $A_1.A_2$ from top to bottom.

Input	Model	Network	Dimension	Val_{MSE}
F_1	M_1	A_1	$\ell_1 \times [128, 128, 128, 128, 1]$	0.0572
F_2	M_2	A_2	$\ell_1 \times [100, 128, 128, 48, 128, 1]$	0.0575
F_3	M_3	A_1	$\ell_2 \times [64, 128, 64, 128, 1]$	0.0585
F_4	M_4	A_2	$\ell_2 \times [64, 64, 128, 48, 64, 1]$	0.0636

equation except for the second submission.

$$V_{fused} = \sum_{k=1}^{m} \alpha_k V_k, \quad \text{w.r.t} \quad \sum_{k=1}^{m} \alpha_k = 1 \tag{1}$$

where V_k denotes the output of model k. In submission 2, we deployed a Support Vector Regression (SVR) with RBF kernel for the fusion. We adjusted the parameters of ensemble models on the validation set. The components of each model were shown in 4. Based on the results from the

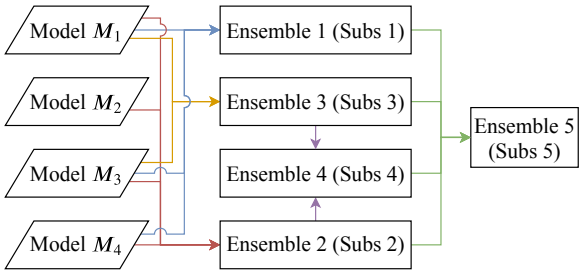

Figure 4: The components of our ensemble models.

validation set, as shown in 2, we conducted the first three submissions with the ensemble of two, three, and four single models. With the difference in the feature representation, the combination of two A_1 got 696×10^{-4}, the best MSE on these submissions. The class-wise MSE has a large gap between the level 0 and the others caused by the imbalance in the development set. At the 4th attempt, we made the ensemble of results from submission second and third, which created by different strategies. They improved the overall MSE as well as the error on level 0. In the final submission, we fused all the previous submission to obtain the final result which achieved the best result with the MSE of 0.0597 and the small gap between class-wise MSE. 3 summarized our results of five submissions in the validation set and test set. The MSE for each engagement level also provided including *disengaged* (DE, 0), *barely engaged* (BE, 1), *engaged* (E,2) and *highly engaged* (HE, 3).

Table 3: The details of submissions.

Subs	Val_{MSE}	Test_{MSE}				
	Overall	DE	BE	E	HE	Overall
1	0.0497	0.3342	0.0834	**0.0133**	0.0660	0.0787
2	**0.0372**	0.3289	0.1087	0.0270	**0.0353**	0.0911
3	0.0518	0.2686	0.0644	**0.0231**	0.0640	0.0696
4	0.0777	**0.2204**	0.0405	0.0320	0.1022	0.0628
5	0.0958	0.2461	**0.0297**	0.0224	0.1378	**0.0597**

4 CONCLUSION

In this paper, we presented a method for engagement prediction "in the wild" environment, a sub challenge in 7th Emotion Recognition in the Wild Challenge 2019. Our approach takes advantage of the state-of-the-art architectures as well as frameworks in computer vision to solve the problem. Our results demonstrate a strong effect of the facial feature as well as eye gaze related features in engagement prediction. With these effective, we got the best performance in the challenge with MSE of 0.0597. Besides that, future research could be continued to explore in the other fields such as engagement in health care, social interaction.

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